

Mall Customer Analysis Report

Data Analytics & Customer Segmentation Project

1 Executive Summary

This project focuses on analyzing mall customer data to understand purchasing behavior, income distribution, and spending patterns. The goal is to identify key customer segments that can help businesses design targeted marketing strategies, improve customer retention, and maximize revenue.

The analysis was performed using Python (Google Colab), MySQL for data storage and querying, and a BI tool (Power BI/Tableau) for dashboard visualization.

2 Problem Statement

Retail businesses often struggle to understand their customers effectively. Without proper segmentation, marketing efforts become inefficient and revenue opportunities are lost.

This project aims to answer:

- Who are the high-value customers?
- How do income and age affect spending behavior?
- Which customer segments should be targeted for marketing campaigns?
- What patterns exist in customer spending behavior?

3 Dataset Overview

The dataset contains mall customer information including:

- Customer ID
- Gender
- Age
- Annual Income (k\$)
- Spending Score (1–100)
- Derived Features:
 - Age Group
 - Spending Score Category

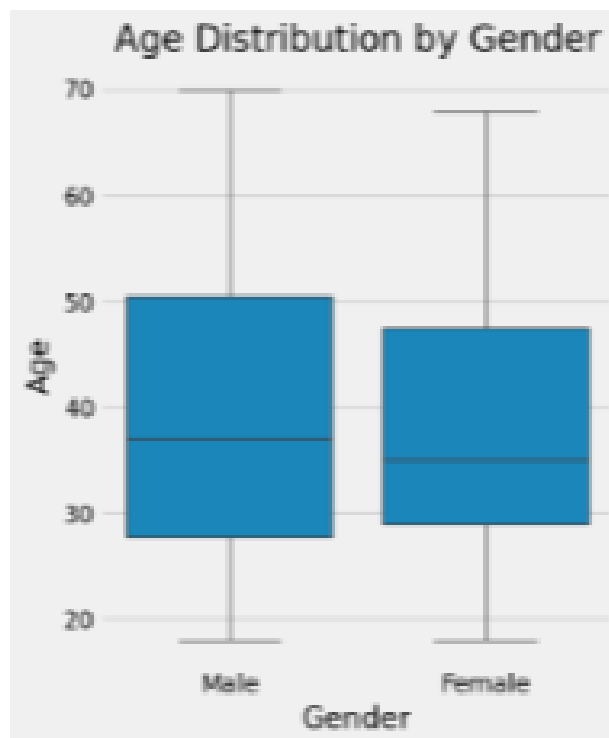
4 Tools & Technologies Used

- Python (Google Colab) – Data Cleaning & EDA
- MySQL – Data Storage & Querying
- SQL – Business Insights Extraction
- Power BI – Dashboard Visualization
- Excel – Supporting Analysis

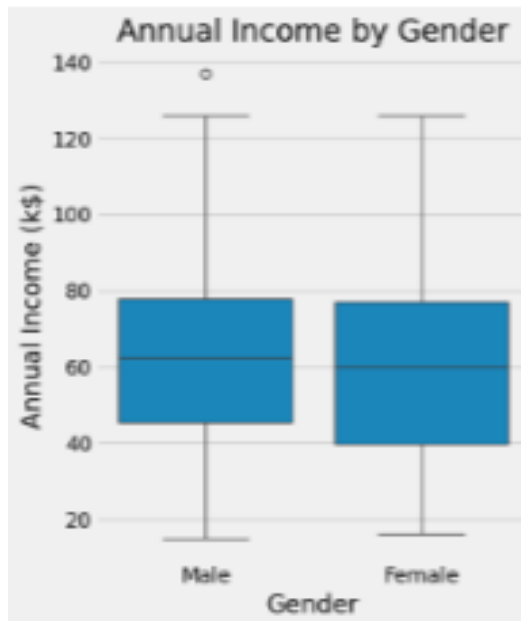
5 Methodology

Step 1: Data Cleaning & EDA (Python)

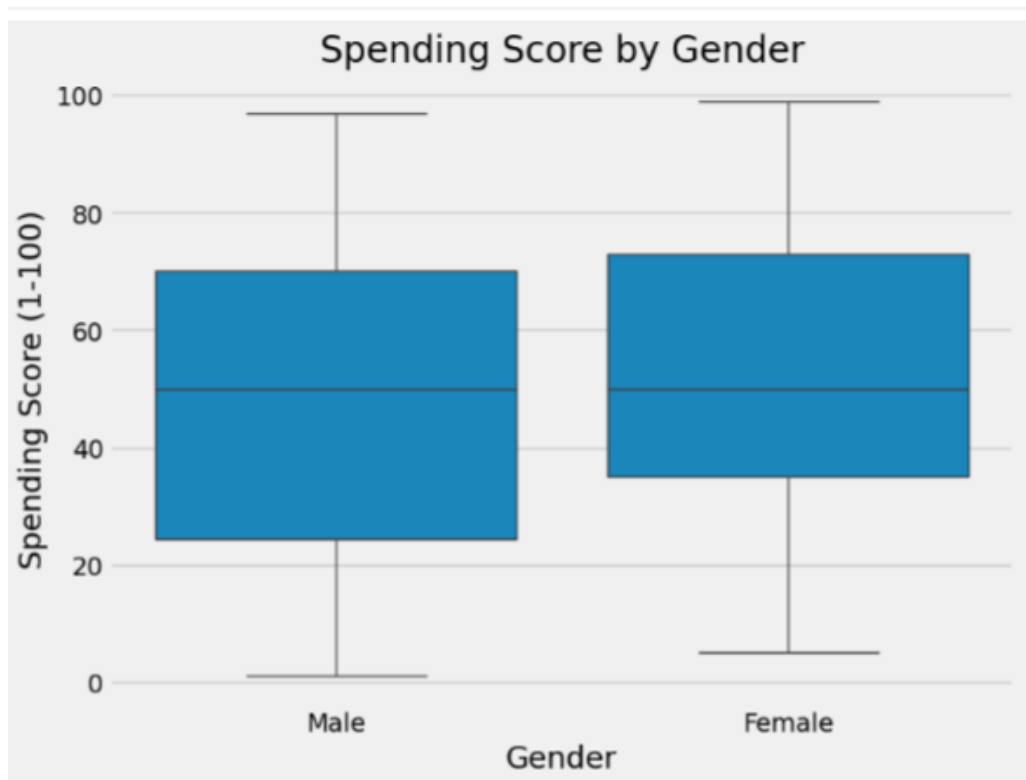
- Handled missing values and duplicates
- Verified data types and consistency
- Performed exploratory data analysis:
 - Age distribution



- Income distribution



- Spending behavior patterns



Step 2: Data Storage (MySQL)

- Created structured database schema
- Loaded cleaned dataset into MySQL

- Verified data integrity using SQL queries

Step 3: SQL Analysis

Performed advanced business queries:

- Customer segmentation by income and spending

CustomerID	Gender	Age	Annual Income	Spending Score	Customer_Segment
1	Male	19	15	39	Low Engagement Customers
2	Male	21	15	81	General Customers
3	Female	20	16	6	Low Engagement Customers
4	Female	23	16	77	General Customers
5	Female	31	17	40	General Customers
6	Female	22	17	76	General Customers
7	Female	35	18	6	Low Engagement Customers

- Identify Top Spending Groups

Spending Score Category	Total_Customers	Avg_Spending
Very High	30	89.7667
High	35	71.3429
Medium	76	49.2237
Low	25	30.2400
Very Low	34	10.3824

- Gender contribution analysis

Gender	Total_Customers	Avg_Income	Avg_Spending_Score
Male	88	62.2273	48.5114
Female	112	59.2500	51.5268

- High-Value Customer Identification

CustomerID	Gender	Age	Annual Income	Spending Score
186	Male	30	99	97
146	Male	28	77	97
128	Male	40	71	95
168	Female	33	86	95
142	Male	32	75	93
164	Female	31	81	93
174	Male	36	87	92

Step 4: Dashboard Development

Built an interactive dashboard with KPIs and visuals:



Key KPIs:

- Total Customers
- Average Age
- Average Income
- Average Spending Score
- High-Value Customers

Visualizations:

- Gender Distribution (Donut Chart)
- Age Group Analysis (Bar Chart)
- Income vs Spending Trend (Line Chart)
- Spending behavior by segment (Stacked Bar Chart)
- Customer Segmentation Clusters
- Top Customers Table

6 Key Insights

1. Customer Segmentation

Customers are grouped into distinct behavioral segments:

- High Income – High Spending (Premium Customers)
- High Income – Low Spending (Potential Target Group)
- Low Income – High Spending (Value Seekers)
- Low Income – Low Spending (Low Priority Segment)

2. Spending Behavior

- Spending behavior is not strictly dependent on income
- Certain low-income groups show high engagement and spending behavior

3. Age Analysis

- Younger customers show higher variability in spending
- Middle-aged groups show stable purchasing patterns

4. Gender Insights

- Spending patterns are relatively balanced across genders
- Slight variations exist in high-value segments

7 Business Recommendations

- Focus marketing campaigns on high-income low-spending customers
- Introduce loyalty programs for high-value customers
- Create personalized offers based on age groups
- Use targeted ads for underperforming segments
- Enhance customer retention strategies for high-spending low-income groups

8 Conclusion

The Mall Customer Analysis project successfully provides deep insights into customer behavior using data-driven techniques. By combining SQL analysis and interactive dashboards, businesses can make informed decisions to improve customer targeting, increase engagement, and boost revenue.

This project demonstrates strong skills in:

- Data Analysis
- SQL Querying
- Business Intelligence
- Data Visualization
- Customer Segmentation